Energy from Fuels

Aim: to investigate the energy released from burning fuels.

Equipment:

- spirit burners containing paraffin and ethanol;
- small conical flasks;
- stands and clamps;
- heat-proof mats;
- splints;
- thermometers;
- stop clocks;
- goggles;
- measuring cylinders.

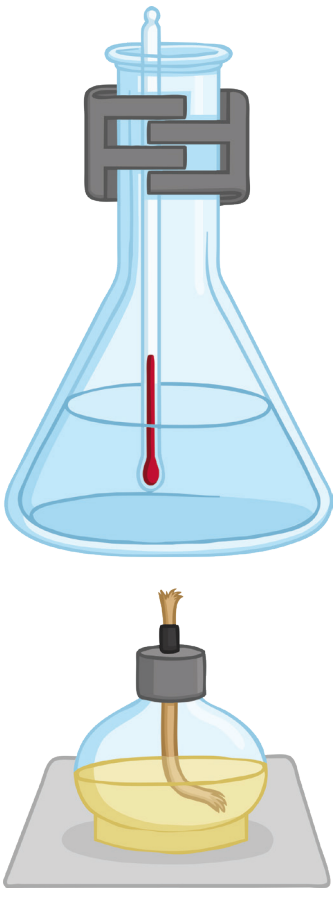
Safety:



- Wear goggles at all times.
- Paraffin and ethanol are very flammable.
- Ensure you know how to use the spirit burners properly.
- Test tubes and water will become very hot after heating.
- Stand up whilst completing the experiment.

Method

1. Set up the equipment as shown at right:
2. Using the measuring cylinder, measure 100cm³ of tap water and pour into the conical flask.
3. Record the temperature of the water.
4. Place the spirit burner containing paraffin under the conical flask and light.
5. Record the temperature of the water every minute until it reaches a temperature of 50ºC, or for no longer than 5 minutes. Replace the cap on the spirit burner.
6. Repeat the experiment using fresh water, and this time using the spirit burner containing ethanol.
7. Remember to record your results on the tables below.



BEYOND SCIENCE

Aim: to investigate the energy released from burning fuels.

Variables:

independent variable: _____

dependent variable: _____

control variables: _____

Results:

Paraffin:

Time (min)	Temperature ºC
1	
2	
3	
4	
5	

Ethanol:

Time (min)	Temperature ºC
1	
2	
3	
4	
5	

Once you have completed the experiment, please present your results in a line graph. You will need to collect a piece of graph paper from the teacher.

If you are unsure how to do this, ask your teacher for the graph help sheet.

Once you have drawn your graph, please complete the tasks below:

Conclusion:

The _____ released the most amount of heat energy. I know this because _

_____

_____

Evaluation:

Most experiments have some errors, or encounter some problems. What was the biggest error, or problem, with your experiment and how would you overcome it next time?

_____

_____

_____

_____

_____